

100 Days of Linux - Day 056

Title: Exploring Network Configuration with ip

Today's Goal

Learn how to inspect network interfaces, IP addresses, and routing information using the **ip** command.

Why This Matters

The **ip** command is the modern replacement for older networking tools such as **ifconfig**. Linux administrators, cloud engineers, and data engineers use it daily to troubleshoot connectivity.

Command of the Day: ip

Common Syntax:

```
ip addr
ip link
ip route
ip -br addr
```

Useful Options:

```
addr = Show IP addresses
link = Show network interfaces
route = Show routing table
-br = Brief output
```

Important Fields:

Interface name, state, IPv4 address, IPv6 address

Beginner Lesson

Every Linux computer has one or more network interfaces. The **ip** command lets you view their status, assigned IP addresses, and how your computer reaches other networks.

Practice Questions

1. Run **ip addr** and identify your primary network interface.
2. Run **ip -br addr** and compare the output with **ip addr**.
3. Run **ip link** and identify which interfaces are UP.
4. Run **ip route** and identify the default gateway if one exists.
5. Display your current working directory after completing today's tasks.

Hints

WSL may show virtual network interfaces. Focus on understanding the command output rather than memorizing every field.

Mini Challenge

Record the interface name, IPv4 address, and default route of your system in a text file named `network_report.txt` using

information from the ip commands.

Common Mistakes

Confusing the loopback interface (lo) with your main network interface. The loopback address 127.0.0.1 is used only by your local machine.

Did You Know?

The ip command is part of the iproute2 package and has largely replaced ifconfig on modern Linux distributions.

Pro Tip

Use ip -br addr when you want a quick, easy-to-read summary of your network configuration.

Answer Key (Instructor Only)

Q1: ip addr

Q2: ip -br addr

Q3: ip link

Q4: ip route

Q5: pwd

Homework

Run the ip commands several times while connecting or disconnecting from different networks (if possible) and observe how the output changes.

Next Day Preview

Tomorrow you'll complete the Networking module with a practical assignment combining ping, curl, wget, ssh, scp, and ip.